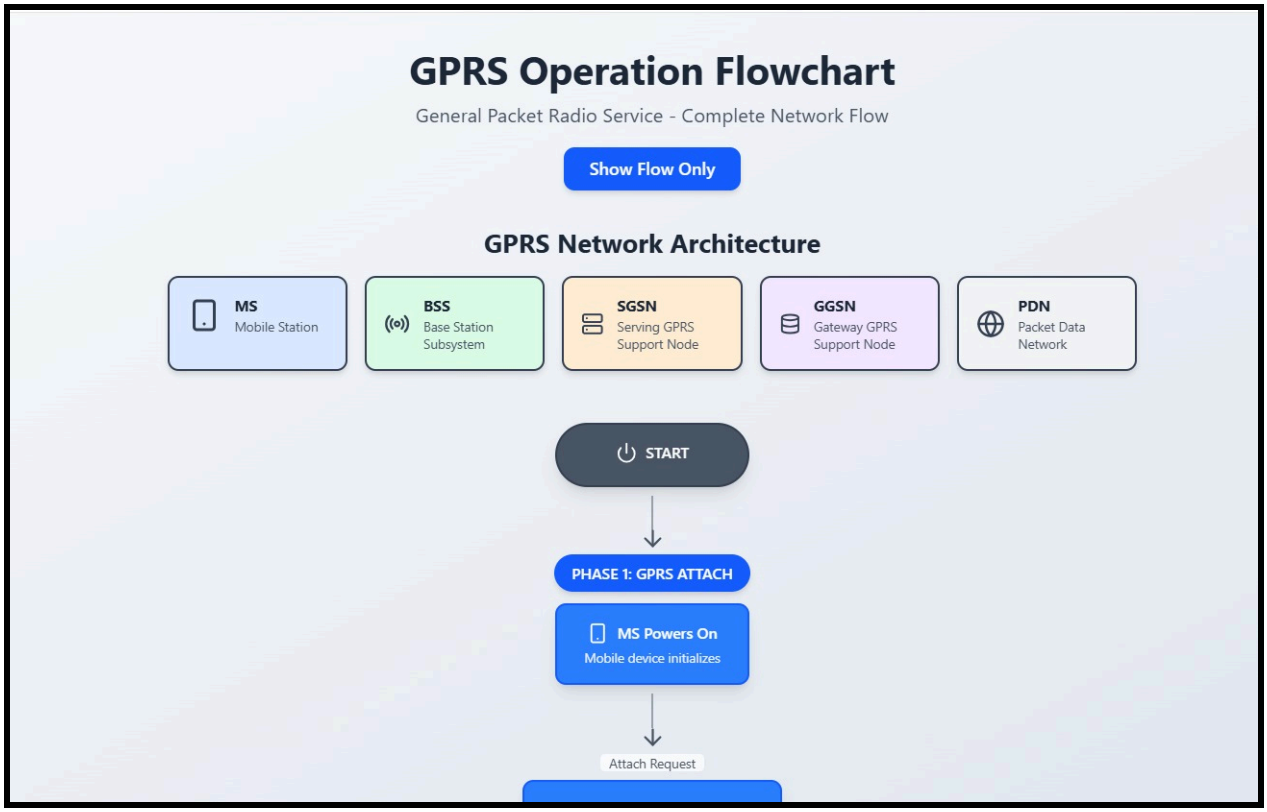
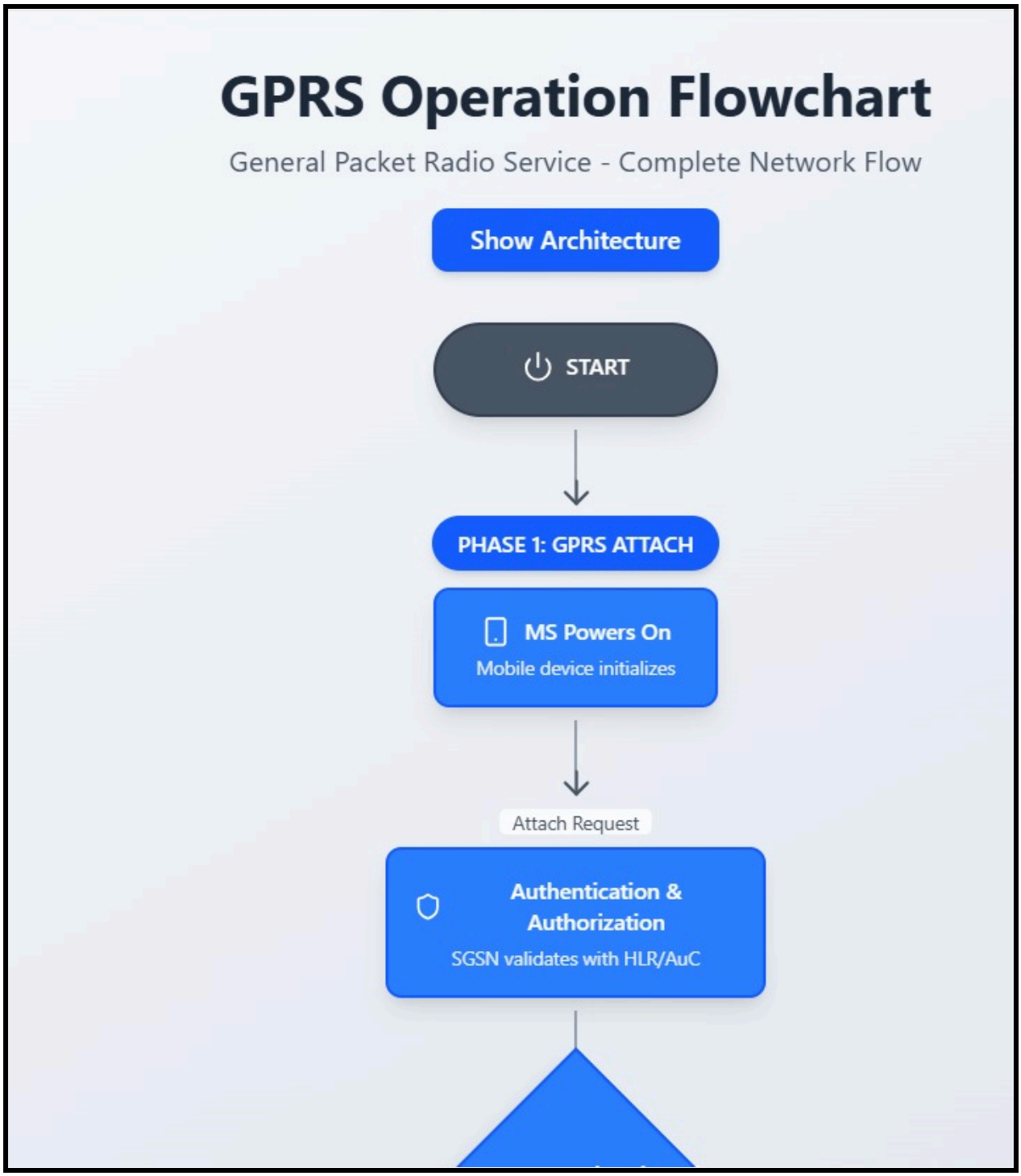
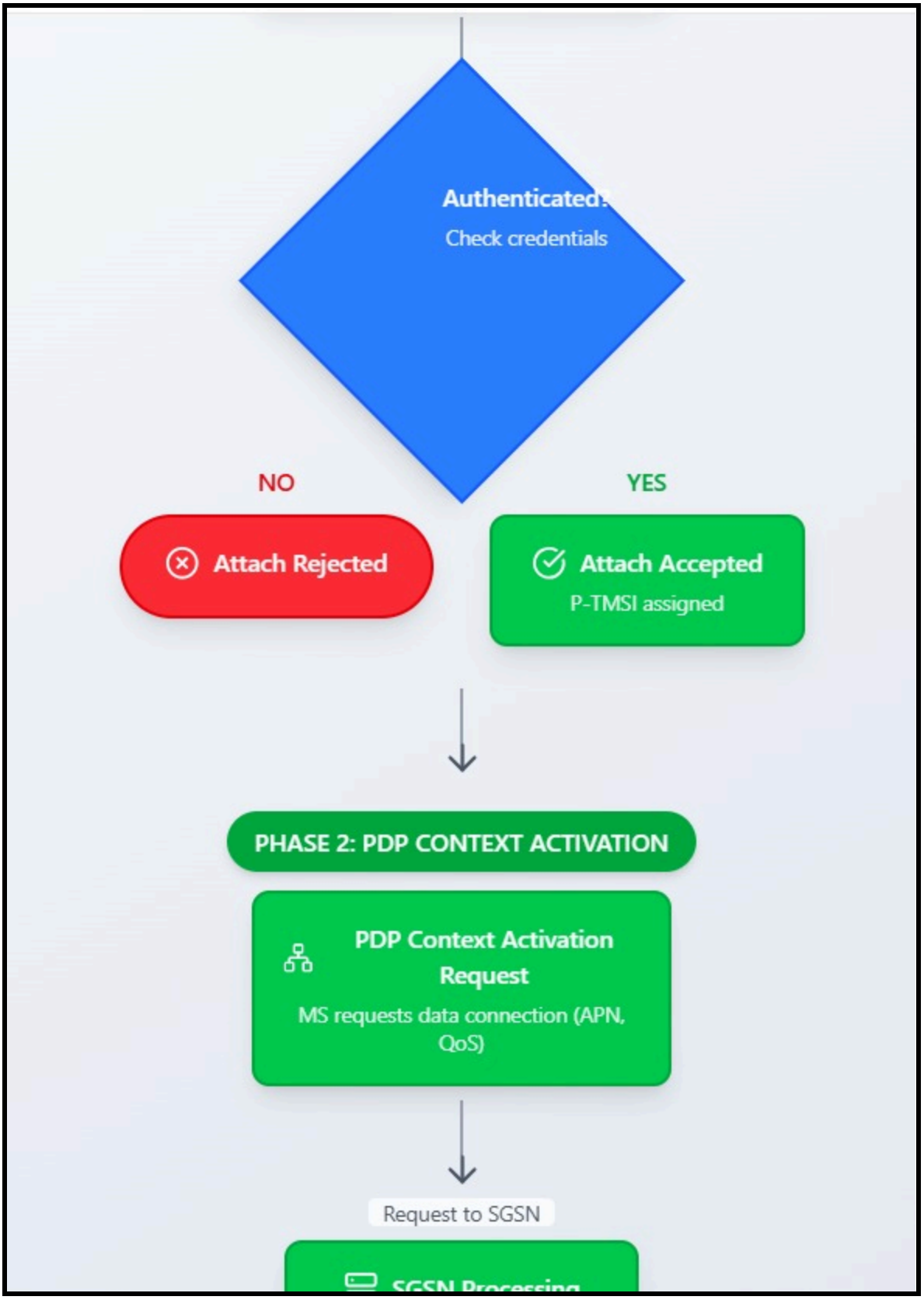
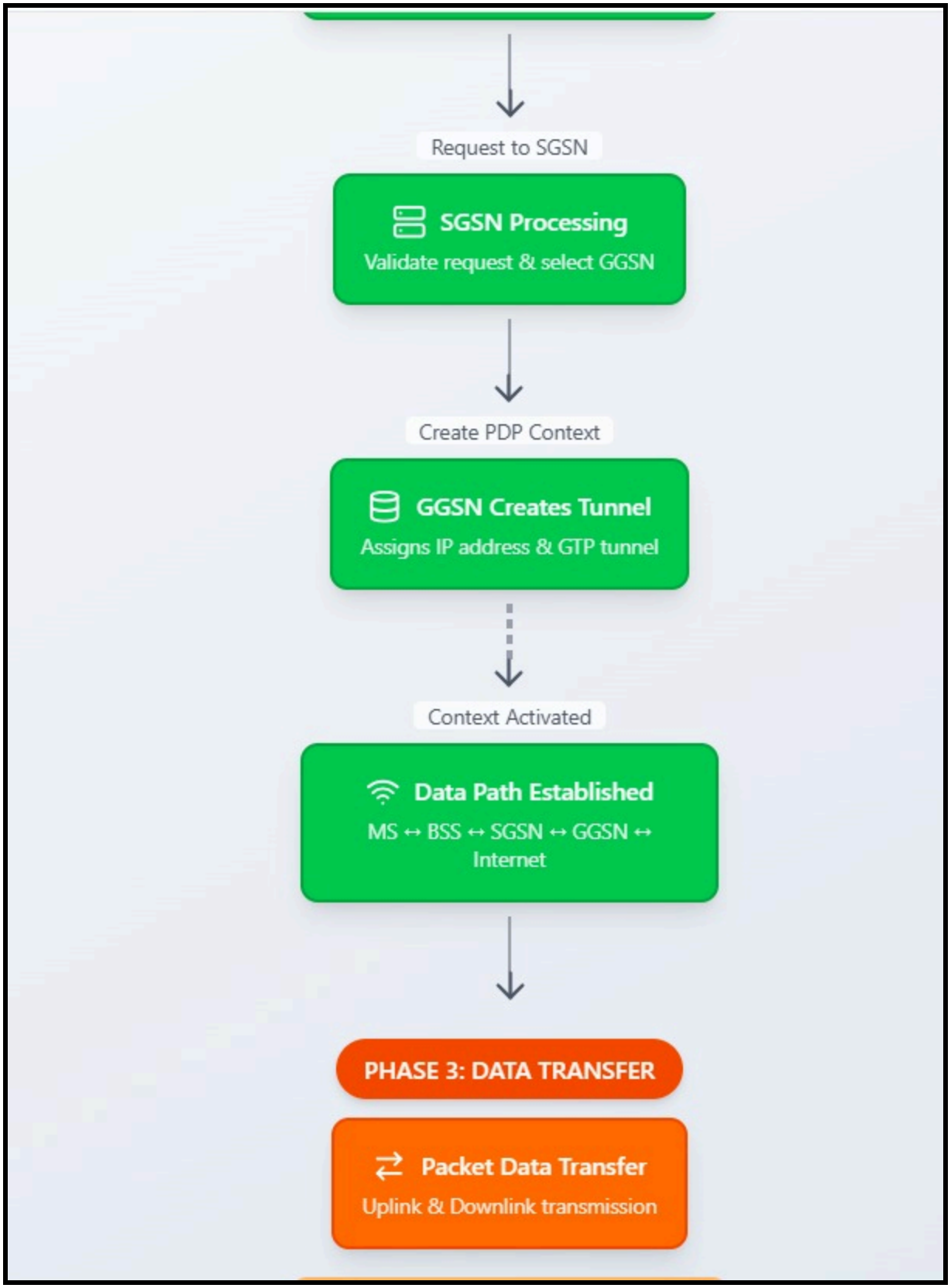
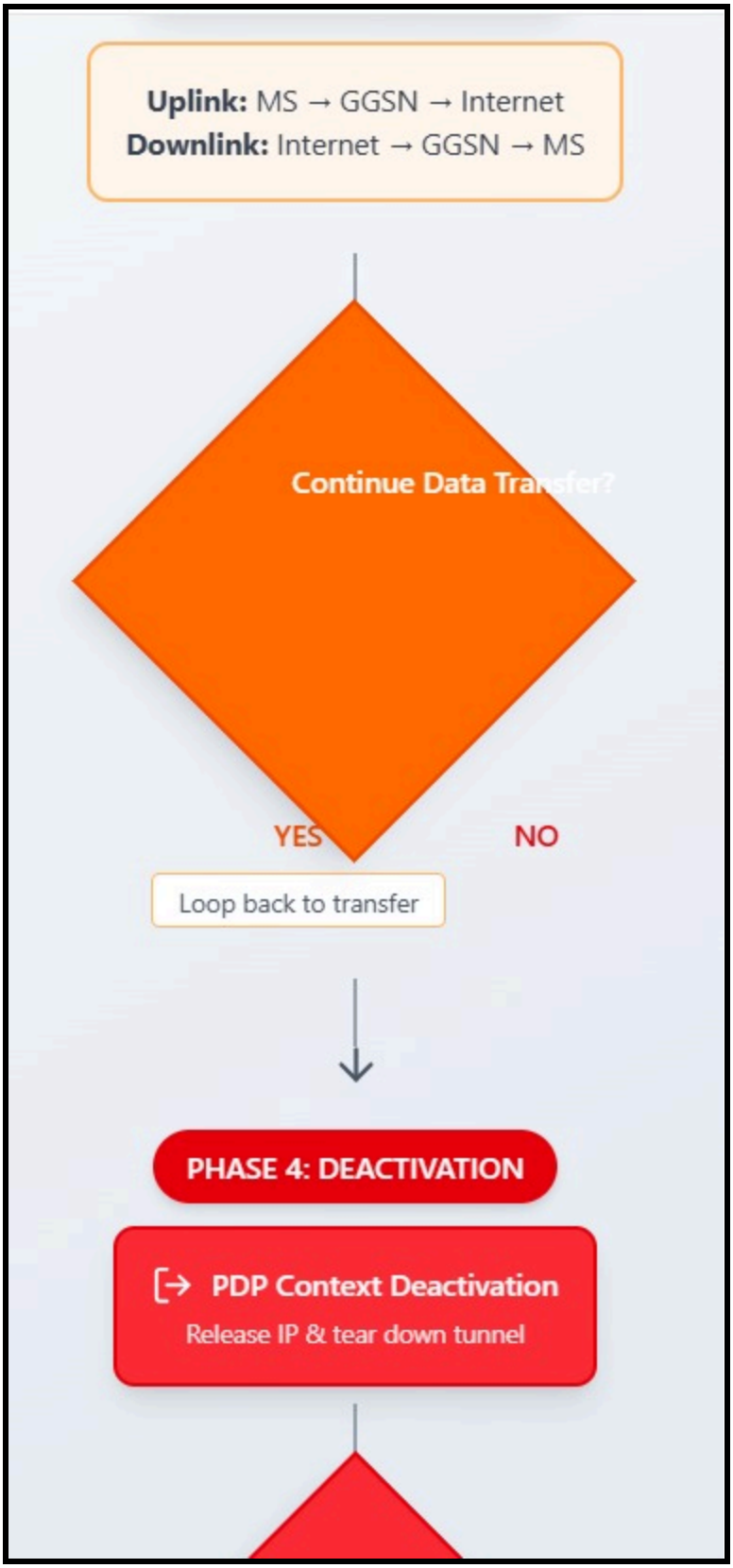
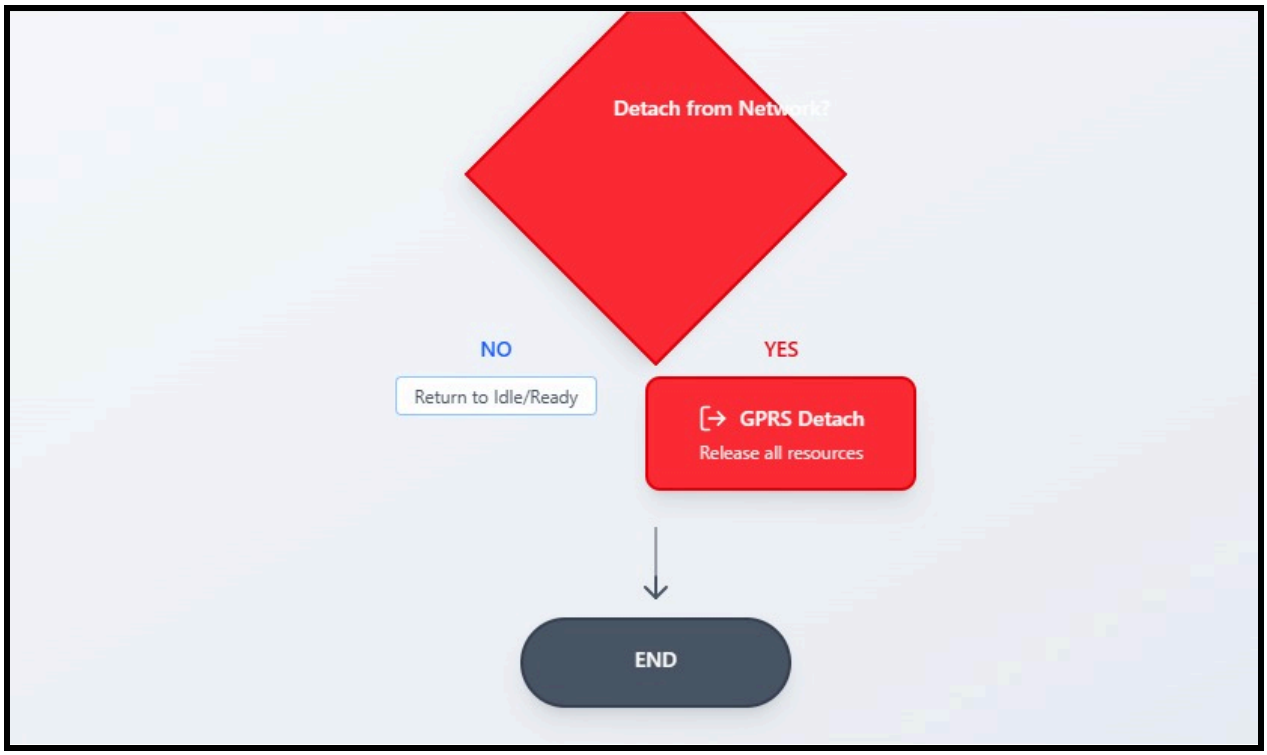

GPRS Operation Flowchart
General Packet Radio Service - Complete Network Flow
Show Flow Only
GPRS Network Architecture
MS
Mobile Station
BSS
Base Station Subsystem
SGSN
Serving GPRS Support Node
GGSN
Gateway GPRS Support Node
PDN
Packet Data Network
START
PHASE 1: GPRS ATTACH
MS Powers On
Mobile device initializes
Attach Request


GPRS Operation Flowchart
General Packet Radio Service - Complete Network Flow
Show Architecture
START
PHASE 1: GPRS ATTACH
MS Powers On
Mobile device initializes
Attach Request
Authentication & Authorization
SGSN validates with HLR/AuC


Authenticated?
Check credentials
NO
YES
Attach Rejected
Attach Accepted
P-TMSI assigned
PHASE 2: PDP CONTEXT ACTIVATION
PDP Context Activation Request
MS requests data connection (APN, QoS)
Request to SGSN


Request to SGSN
SGSN Processing
Validate request & select GGSN
Create PDP Context
GGSN Creates Tunnel
Assigns IP address & GTP tunnel
Context Activated
Data Path Established
MS ↔ BSS ↔ SGSN ↔ GGSN ↔ Internet
PHASE 3: DATA TRANSFER
Packet Data Transfer
Uplink & Downlink transmission


Uplink: MS → GGSN → Internet
Downlink: Internet → GGSN → MS
Continue Data Transfer?
YES
NO
Loop back to transfer
PHASE 4: DEACTIVATION
PDP Context Deactivation
Release IP & tear down tunnel



Key Components & Protocols

MS:
Mobile Station with GPRS capability

BSS:
Radio interface (BTS + BSC)

SGSN:
Mobility management & authentication

GGSN:
Gateway to external networks

P-TMSI:
Packet Temporary Mobile Subscriber Identity

GTP:
GPRS Tunneling Protocol

PDP Context:
Packet Data Protocol session

APN:
Access Point Name